

BACHELOR ECONOMETRICS



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High-Frequency Algorithmic Bitcoin Trading Using Both Financial and Social Features

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Statement of Originality

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I declare that the text and the work presented in this document are original and that no sources other than those mentioned in the text and its references have been used in creating it.

The Faculty of Economics and Business is responsible solely for the supervision of completion of the work, not for the contents.

Abstract

The goal of this research was to study and compare the performance of HFT-algorithms for trading Bitcoins on cryptocurrency exchanges. For that matter, two Random Forest models were created, one with the inclusion of social indicators and one without. It is shown that both these models were profitable on the Bitcoin market. While there is no statistical evidence that the annualized Sharpe ratio of the algorithm with social features outperforms the one without those features, it has been demonstrated that the model with social indicators has significantly higher F1-scores than the one without those indicators. Finally, it has been found that these two HFT-algorithms do not significantly outperform the annualized Sharpe ratio from a simple Buy&Hold strategy, which was in contrast to the findings from past research.

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Introduction

In the current days, it has become a very normal practice to read daily newspaper headlines on cryptocurrencies. In most cases, those newspapers report an immense increase or drop in the value of the Bitcoin. For example, on 18 April 2021, the Bitcoin price dropped from an all-time high of \$60,884.85 to \$52,810.06, meaning a price decrease of almost 15%, in only a few hours ([Menton, 2021](#)). This high volatility in the trading price of Bitcoin has invoked multiple questions among many people about the origin, the speculative nature, and the possible profits when trading this cryptocurrency.

Bitcoin is the first virtual currency in the world that makes decentralized payments possible. It originated in 2008 by an anonymous white paper by [Nakamoto \(2008\)](#). This anonymous author (or group) invented the first digitalized anonymous payment recording system called the blockchain. This system makes sure that the payments are processed securely so that a central authority overseeing the currency is not necessary. With a maximum of 21 million Bitcoins to be reached around the year 2140, the scarcity in the market is presumably one of the reasons why Bitcoin is highly-priced ([Böhme, Christin, Edelman, & Moore, 2015](#)). After the invention of Bitcoin, many more different sorts of cryptocurrencies were created, the most important ones being Ethereum, Ripple, and Litecoin. To compare the popularity of those newer cryptocurrencies to Bitcoin, together those three cryptocurrencies take up approximately 17% of the total cryptocurrency market capitalization, while Bitcoin accounts for 51% of that total market cap ([Coin Market Cap, 2021a](#)).

The trading of those cryptocurrencies is mainly done on the so-called cryptocurrency exchanges ([Fang et al., 2020](#)). These are 24-hour exchanges where cryptocurrencies can be bought or sold for real currencies. For example, Binance is considered to be one of the biggest cryptocurrency exchanges where Bitcoins can be traded for USD. These kinds of cryptocurrency exchanges are also highly suitable for algorithmic trading strategies, meaning the usage of computer-based algorithms to perform automatic trades on exchanges. These algorithms are

in most cases based on Machine Learning (ML) techniques. Usually, this algorithmic trading is performed on a very short time basis, called the trading horizon, which can include intervals as narrow as (nano)seconds to wider ones covering multiple hours.

However, algorithmic trading performed on those cryptocurrency exchanges could potentially lead to very diverse outcomes. [Glaser, Zimmermann, Haferkorn, Weber, and Siering \(2014\)](#) found that the reason for this finding is that the cryptocurrency market is very speculative. This means that most investors only wish to profit from the market's fluctuations and are less interested in the virtual currency itself. The authors state that this speculative nature is being fueled by the fact that the intrinsic value of Bitcoin is zero. As a consequence, Bitcoin can rapidly increase or decrease in price in a matter of hours or even minutes. Therefore, algorithmic trading strategies that execute trades on a narrow trading horizon have the potential to use the speculative market for their benefit and make profits.

In the last few years, there has been much research done on the profitability of this algorithmic trading. Most of this research is focused on different types of algorithms, ranging from Machine Learning methods like Random Forests to Deep Learning methods like Neural Networks to train these trading models. However, the outcomes of these different trading strategies were widely varying. From a slightly negative annualized return to a positive annualized return of approximately 400%, the differences between different papers were apparent ([Borges & Neves, 2020](#); [Sebastião & Godinho, 2021](#); [Vo & Yost-Bremm, 2020](#)). Note that these contrasting results were found in studies that use Bitcoin data from different periods in time. More about this can be found in the next chapter [Theoretical Framework](#). This means that different algorithmic trading models, as well as different training data sets, affect the profitability of algorithmic trading strategies.

Furthermore, these algorithms are trained on a set of features, where nearly all papers use either financial indicators or Bitcoin prices directly as training features. Financial indicators are features that describe the cryptocurrency market just before the moment of trading. However, only several papers also consider training these algorithms with the inclusion of social indicators, retrieved from sentiment analysis. Therefore, this paper will focus on creating a trading algorithm for Bitcoins while including both financial features as well as social features to train the algorithm.

This paper will start with a [Theoretical Framework](#) to get a deeper understanding of algorithmic trading with Bitcoins and the theory about the inclusion of both financial and social features in the trading algorithm. In chapter [Methodology](#), both the algorithm and the data that will be used in this research will be explained in detail. Then, in chapter [Results](#), the results of the final trading algorithms and their profitability will be covered. After that, the results will be compared and summarized with regards to the other research on trading algorithms for Bitcoins in the chapter [Discussion and Conclusion](#).

Theoretical Framework

In this literature review, high-frequency algorithmic trading (HFT) strategies for cryptocurrencies are studied. To have a clear understanding of those strategies and their profitability on cryptocurrency markets, this literature review will start with the efficiency of the Bitcoin market and its possible profitability. Then a brief explanation about HFT on cryptocurrency markets is given. Consequently, a few different types of HFT will be explained in more depth, then the financial indicators where the trading algorithms are built on will be summarized and lastly, the use of social indicators in HFT will be highlighted.

2.1 Financial Economics

Briefly touched upon in the previous chapter [Introduction](#), cryptocurrency markets are said to be highly speculative ([Glaser et al., 2014](#)). Comparing this to the efficient market hypothesis ([Fama, 1970](#)), a market is thought to be efficient when the prices in the market fully represent all available information. This means that the higher the efficiency of the market, the less easy it is to consistently outperform the market. This then again makes the market less speculative. [Urquhart \(2016\)](#) has found that the Bitcoin market at the moment of that research was inefficient. Thus, this is consistent with the results from [Glaser et al. \(2014\)](#) who have found that the Bitcoin market is highly speculative. There are several reasons why the Bitcoin market was found to be inefficient, the most evident one being the effect of behavioral finance biases like over-optimism, overreaction, and bounded rationality ([Peng & Xiong, 2006](#)). These three biases relate to chasing past success and expecting that success to continue in the future, subconsciously ignoring the uncertainty. The consequence of this all is that the market inefficiency could make it possible to gain excessive profits on trading Bitcoins. This is a good sign for algorithmic-based trading because these kinds of strategies can exploit the speculative nature of markets ([Yadav, 2015](#)).

However, [Urquhart \(2016\)](#) also found that the Bitcoin market might be moving towards efficiency. This finding was confirmed by [Sensoy \(2019\)](#). This researcher observed that the efficiency of algorithmic-based trades has been increasing over the last few years. Next to this, [Mnif, Jarboui, and Mouakhar \(2020\)](#) found that the market efficiency has been increasing even more during the COVID-19 pandemic. All these results combined show that algorithmic trading on the Bitcoin market nowadays could be less profitable than a few years ago when the market was still considered to be inefficient. This means that it is expected that trading algorithms for Bitcoins trained on recent data will be less profitable than results from previous research trained on earlier data.

2.2 High Frequency Trading

As stated by [Vo and Yost-Bremm \(2020\)](#), HFT is defined as “the use of computerized trading algorithms to buy and sell assets quickly and frequently, with a short holding period to earn minuscule margins on each trade” (p. 556). While this high-frequency trading has been widely used on multiple financial markets, the employment of this strategy on cryptocurrency markets is thought of it still being at an early stage ([Böhme et al., 2015](#)). It is for that reason that in the last few years, several studies have been performed on those HFT strategies for cryptocurrencies, and in particular on the profitability of these strategies.

[Vo and Yost-Bremm \(2020\)](#) have built an HFT-model for trading Bitcoins based on the machine learning technique called ‘Random Forest’ (RF). A Random Forest is a kind of classification tree ensemble model that creates multiple decision trees with added randomness ([Breiman, 2001](#)). Their model is based on financial indicators formed by minute data from seven different Bitcoin exchanges from the period 2012-2017. To evaluate the performance of the RF model, the authors compared this model with a Deep Learning model (with a Binary Cross Entropy loss function) and found that the Random Forest outperformed this Deep Learning model. Then, to check the economic performance of the model, the authors also performed an economic value evaluation to simulate how well the model would be able to generate profits while trading Bitcoins. For this reason, [Vo and Yost-Bremm \(2020\)](#) created a rolling, out-of-sample simulation where data from the previous week was used to make trading decisions in that subsequent week. This evaluation model had exceptional results compared to other researches. More specifically, the selected model predicted an annualized Sharpe ratio¹ of 8.22 and an accuracy of 97% for trading frequencies of 15 minutes. This paper thus shows that algorithmic HFT-models on cryptocurrency markets could be very promising under certain conditions. Nevertheless, it is good to emphasize that this accuracy did decline over the years in the sample. Another thing that is good to note is that the Random Forest is trained on one randomly selected Bitcoin exchange,

¹The annualized Sharpe ratio is given as the expected daily return divided by the standard deviation of those daily returns multiplied by $\sqrt{365}$, see Equation 3.2 ([Sebastião & Godinho, 2021](#)).

CoincheckJPY here, and is tested on the six other exchanges that are taken into account in this research. As the behavior of all the Bitcoin exchanges is extremely similar (Böhme et al., 2015), it would come to no surprise that the indicators from one exchange could very well estimate the Bitcoin prices on the other exchanges in that period. This could be a possible explanation for the extremely high accuracy found in this research.

However, different types of those HFT strategies have widely varying and even contradicting outcomes. Sebastião and Godinho (2021) have built an HFT machine learning model that studies ensembles of different kinds of algorithms from past literature (including the RF strategy as studied by Vo and Yost-Bremm (2020)). This research considered daily data on Bitcoin, Ethereum, and Litecoin for the period from 2015 to 2017. Sebastião and Godinho (2021), in contrast to Vo and Yost-Bremm (2020), found that most of the ensembles of the algorithms had a low or even a negative annualized Sharpe ratio compared to Vo and Yost-Bremm. They found that the maximum annualized Sharpe ratio of 0.95 was reached for trading Ethereum. The authors argued that this lower annualized Sharpe ratio could be caused by a different trading horizon: Vo and Yost-Bremm used a 15-minute interval while Sebastião and Godinho used daily trading. Another possibility that was argued was the difference in trading periods: Vo and Yost-Bremm used 5 years in which the price of the Bitcoin rose immensely, while Sebastião and Godinho used a more narrow period in which the price of the Bitcoin dropped (CoinDesk, 2021). These results have shown that the trading frequency and the trading period considered in the research are important to the outcomes of the machine learning models. This is consistent with the theory from the last subsection [Financial Economics](#). There it was argued that a more efficient Bitcoin market could be the cause of less excessive profits and hence could be the cause of a lower Sharpe ratio. Concluding, this subsection has shown multiple outcomes of HFT algorithms for trading Bitcoins. This means that it is expected that both the chosen machine learning model and the trading frequency of the model affect the profitability of an HFT strategy on Bitcoins.

2.3 Indicators

Additionally, including different kinds of features in the algorithms could cause a significant change in their performance. For example, Borges and Neves (2020) also studied an ensemble of different machine learning models (including RF, Linear Regression, and Support Vector Machines) on resampled 1-minute Bitcoin data. These authors found a trading strategy with a maximum annualized Sharpe ratio of 1.5 and an accuracy up to 59%. In that research, the machine learning models were constructed with nine different financial indicators. An example of such a financial indicator is the Relative Strength Index (RSI) that measures the speed and change of price movements (Borges & Neves, 2020). Huang, Huang, and Ni (2019) have extended this idea of including multiple financial indicators. These authors focused on designing an algo-

rithm that is constructed on 124 distinct financial indicators. In the paper, a Random Forest is trained on daily Bitcoin data from 2012 to 2017. Note that all the financial indicators are added to the model separately, which could have caused multicollinearity issues. It is shown that the proposed Random Forest model has strong predictive strength for daily returns of Bitcoin. However, both an overall accuracy score and an annualized Sharpe ratio of the strategy are not reported. This makes it difficult to compare the results of the paper to similar literature.

[Z. Chen, Li, and Sun \(2020\)](#) studied the predictability of future (5-minute interval) Bitcoin prices and found an accuracy reaching 67%. This research is based on both financial and social indicators. Note that in this context, the attention to Bitcoins both online and in the media is meant by the social indicators. Although [Z. Chen et al. \(2020\)](#) did not use these predicted prices to create an HFT-strategy, these results did show that including social features could potentially ensure higher accuracy in such a strategy.

Even though there has been found evidence that the annual returns of Bitcoin could be predicted by Google searches for ‘Bitcoin’ ([Aalborg, Molnár, & de Vries, 2019](#)), the research considering the inclusion of social features in high-frequency trading strategies for cryptocurrencies has been sparse. Alike the study earlier mentioned by [Z. Chen et al. \(2020\)](#), there have been performed several analyses on the predictability of future prices of Bitcoin using social features, but not on HFT-strategies that consider these social indicators. Do note that a high accuracy in these prediction models is very likely to assure a high profitability in trading models ([Ertimur, Sunder, & Sunder, 2007](#)). [Garcia and Schweitzer \(2015\)](#), one of the very few that did study this matter, found an annualized Sharpe ratio of 1.8 for trading Bitcoins on daily basis in the period 2011-2014. This showed that even in a period before Bitcoin spectacularly rose in value ([CoinDesk, 2021](#)), including social features in HFT-algorithms increased the profitability of the strategy. Hence, it could be interesting to research whether the inclusion of social features in HFT-strategies still could cause an increase in profitability of these strategies.

Concluding, in this subsection, there is found evidence that various types of both financial and social indicators could affect the results from algorithmic trading models. Subsequently, it is expected that including both financial and social indicators in an HFT model will induce higher profitability of that model than including only financial indicators.

Methodology

In this chapter Methodology, the HFT algorithm that was constructed for this research will be described in detail. Firstly, the data where this algorithm is built upon will be presented. Subsequently, the algorithm itself will be explained in depth.

3.1 Data

In this paper, minute-based trading data is obtained from the Bitcoin trading exchange Binance. The trading data is scraped in Python using the API of this exchange ([Binance, 2021](#)). Note that this exchange is chosen because it is likely to represent the Bitcoin market well, as Binance is in the top 5 of largest cryptocurrency exchanges in the world ([Coin Market Cap, 2021b](#)). This dataset contains the elements time of the trades (at minute level), the volume of Bitcoins traded, and the corresponding Open-High-Low-Close prices. On this exchange, Bitcoins are traded for USD. The trading period that is considered in this research is from January 1, 2021, to April 29, 2021. In that period, the price of the Bitcoin has risen immensely, from \$29,111.52 on January 1 compared to \$53,540.48 on April 29. The course of Bitcoin prices over time can be found in Figure 3.1. This graph shows that the Bitcoin price indeed has risen over time, but also has significantly dropped at the end of the sample period. The descriptive statistics of the Bitcoin prices on the cryptocurrency exchange Binance are summarised in Table 3.1. Note that the difference between the minimum and maximum Bitcoin closing price is very large. Lastly, the trading fees of trading on Binance are 0.1% ([Stone, 2021](#)).

Exchange	Sample period	N	Mean	SD	Max	Min	Median
<i>Binance</i>	01/01/2021-29/04/2021	170757	47977.8	9967.2	64800.0	28241.9	50022.2

Table 3.1: Descriptive statistics of the Bitcoin closing prices at the cryptocurrency exchange Binance.

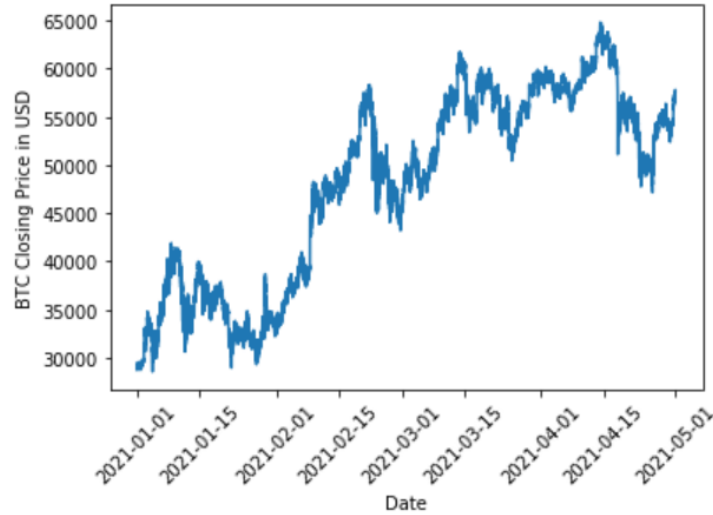


Figure 3.1: The daily closing prices of the Bitcoin for the period from January 1, 2021, to April 29, 2021 on the cryptocurrency exchange Binance.

Next to this, Twitter and Google Trend data are collected separately. Google Trend data is scraped by accessing the Google Trend dataset ([Google, 2021](#)) for the search term 'Bitcoin'. This dataset consists of daily data on the relative number of searches on Google containing the word 'Bitcoin', where the day with the most searches is set to be equal to 100. This dataset thus consists of 119 observations from January 1, 2021, to April 29, 2021. The Twitter dataset is collected with the Augmento API ([Augmento, 2021](#)). This dataset consists of 93 distinct sentiment analysis indicators, constructed by the Augmento API out of Twitter data on #bitcoin. These indicators have been recorded on a minute basis in the period between January 1, 2021, and April 29, 2021. All these indicators consist of the number of tweets in the 1-minute interval that are associated with that particular social indicator. As an example, one of these indicators 'Price' is plotted in Figure B.2 in Appendix B. Both in the next subsection [Methods](#) as in Appendix A, the sentiment analysis indicators that are considered to be used in this research will be described in more detail.

3.2 Methods

In order to construct a profitable HFT algorithm for trading Bitcoins, this subsection will delve into the models that will be utilized in this paper. This means that this subsection will cover the hypotheses that are tested, the models that are created to research these hypotheses, and the features that are used to train these models.

First of all, as already stated in the [Introduction](#), the goal of this research was to create a profitable high-frequency trading model to trade Bitcoins. This model is created by picking the best performing machine learning model and by choosing the set of features that will best predict

a positive return on Bitcoins. Note that the best performing model in this paper is defined as the most profitable trading strategy. A trading strategy is said to be profitable when the returns on that strategy are positive. In terms of machine learning models, the model is thought to be profitable when it can accurately predict if the Bitcoin price is going to increase or decrease in the next period. The more accurate the machine learning model will be able to predict this decision, the more accurate the decision to buy or sell Bitcoins in the next period will be, and hence the higher the possible chance of making profit (Ertimur et al., 2007). This means that the goal of this research is to create an accurate machine learning model that models the decision to buy or sell Bitcoins. However, it has to be noted that a trading strategy like this does not take into account the expected amount that the Bitcoin price will increase or decrease, which could also be a factor that could influence the possible profitability of trading strategies.

The past chapter [Theoretical Framework](#) leads to a few hypotheses for this HFT model on Bitcoins. First of all, the subsection [Financial Economics](#) has shown that it is expected that algorithmic trading on Bitcoins with recent data will be less profitable than results from previous research based on less recent data. In terms of the machine learning model, it is expected that the accuracy of predicting the Bitcoin price in the next period will be lower than the accuracy in earlier research. For example, the accuracy of 97% that Vo and Yost-Bremm (2020) found is expected not to be reached in this research. On the other hand, in previous research, it has been shown that an HFT model on trading Bitcoins was still more profitable than the Buy & Hold strategy, even with recent data (Fang et al., 2020). Therefore, it is expected that the profitability of the HFT model will outperform the Buy & Hold strategy. A measure that is widely used in the determination of the profitability of HFT strategies is the annualized Sharpe ratio (S_a) as touched upon in the [Theoretical Framework](#). This annualized Sharpe ratio is defined as follows:

$$S_{a,r} = \frac{\mu_r}{\sigma_r} \cdot \sqrt{365},$$

where r represents the daily returns of a certain trading strategy, μ_r the expected value of that strategy and σ_r the standard deviation of that strategy. This then leads to the first hypothesis,

$$H_0 : S_{a,HFT} = S_{a,B\&H} \quad \text{vs.} \quad H_1 : S_{a,HFT} > S_{a,B\&H}. \quad (3.1)$$

This hypothesis will be tested with the approach as defined by Jobson and Korkie (1981) and improved by Memmel (2003). This test is a specifically constructed to test the difference between two Sharpe ratios and is defined as follows. For this matter, let $\hat{\mu}_H$ and $\hat{\mu}_B$ represent the means of the excess returns of the HFT and the Buy&Hold strategies respectively and $\hat{\sigma}_H$ and $\hat{\sigma}_B$ the standard deviations of these two strategies. Then let $\hat{\sigma}_{H,B}$ represent the covariance between the two strategies. Lastly, let n represent the number of observations. The Jobson and Korkie (1981) test statistic $\hat{z}$ is then given by

$$\hat{z} = \frac{\hat{\sigma}_B \hat{\mu}_H - \hat{\sigma}_H \hat{\mu}_B}{\sqrt{\hat{\theta}}},$$

in which

$$\hat{\theta} = \frac{1}{n} (2\hat{\sigma}_H^2 \hat{\sigma}_B^2 - 2\hat{\sigma}_H \hat{\sigma}_B \hat{\sigma}_{H,B} + \frac{1}{2}\hat{\mu}_H^2 \hat{\sigma}_B^2 + \frac{1}{2}\hat{\mu}_B^2 \hat{\sigma}_H^2 - \frac{\hat{\mu}_H \hat{\mu}_B}{\hat{\sigma}_H \hat{\sigma}_B} \hat{\sigma}_{H,B}^2).$$

This test statistic $\hat{z}$ has been proven to be asymptotically standard normal distributed.

Subsequently, the second subsection [High Frequency Trading](#) of the [Theoretical Framework](#) has shown that both the chosen machine learning model and the trading frequency of the model could affect the profitability of an HFT strategy on Bitcoins. In that same subsection, it became clear that there were accurate and profitable results found with Random Forest models ([Fang et al., 2020](#); [Sebastião & Godinho, 2021](#); [Vo & Yost-Bremm, 2020](#)). Therefore, it is chosen to construct a Random Forest model in this research. As explained in the last chapter [Theoretical Framework](#), a Random Forest model is a machine learning model that constructs an ensemble of decision trees by bootstrap aggregation (bagging). The Random Forest in this research is based on the minute-based Bitcoin data from the exchange Binance as described in the subsection [Data](#). This data is split into a training period from January 1, 2021, to March 31, 2021, and a test period from April 1, 2021, to April 29, 2021, such that the training sample is approximately three times as large as the test sample. The Random Forest model itself is built in Python using the Scikit Learn package for Random Forest Classifiers ([Pedregosa et al., 2011](#)). It is then validated with 10-fold cross-validation. The optimal number of trees in the Random Forest model is found by testing a range from 200 to 1000 trees. The feature set on which this Random Forest model is trained is described further on in this subsection. This Random Forest then predicts whether the price of Bitcoin has increased or decreased in the next period. It is expected that a trading frequency of 5 minutes will give the highest accuracy for trading Bitcoins ([Borges & Neves, 2020](#)). The model is therefore trained with trading frequencies of 5 minutes.

The last subsection [Indicators](#) of the chapter [Theoretical Framework](#) showed that the inclusion of different sorts of indicators could also affect the profitability of an HFT strategy. Therefore, it is expected that including both financial and social indicators in an HFT model will induce higher profitability of that model than including only financial indicators. In this paper, the accuracy of machine learning models is measured with the F1-score, which is defined by the following equation ([Van Rijsbergen, 1974](#)):

$$F1 = \frac{2}{\frac{1}{precision} + \frac{1}{recall}} = \frac{2}{\frac{TP+FP}{TP} + \frac{TP+FN}{TP}},$$

where TP stands for True Positives (correctly predicting Bitcoin price increases), FP for False Positives (falsely predicting an increase while the Bitcoin price decreased), and FN for False Negatives (falsely predicting a decrease in Bitcoin price). In mathematical notation, the two

hypotheses that will therefore be tested are given as follows:

$$H_0 : F1_{fin+soc} = F1_{fin} \quad \text{vs.} \quad H_1 : F1_{fin+soc} > F1_{fin}, \quad (3.2)$$

$$H_0 : S_{a,fin+soc} = S_{a,fin} \quad \text{vs.} \quad H_1 : S_{a,fin+soc} > S_{a,fin}. \quad (3.3)$$

Hypothesis 3.2 is tested with the Wilcoxon signed ranks test (Demšar, 2006). This test is a non-parametric test to compare two related samples. For this matter, the F1-scores of both the Random Forest model with as well as the one without the social indicators are gathered per day and tested using the Scikit Stats Package 'Wilcoxon' (SciPy, 2021). Hypothesis 3.3 is again tested with the Jobson and Korkie (1981) test as defined before. The indicators themselves are described in detail in Appendix A. Note that the selection of these indicators is based on past research (Borges & Neves, 2020; Z. Chen et al., 2020). In other words, it is chosen to include the same nine financial indicators that Borges and Neves (2020) have used.

Alike the research of Z. Chen et al. (2020), in this paper it is chosen to test the performance in terms of profitability of the trading strategies while possibly including social indicators based on a range of indicators that are described in the Appendix Indicators. Because the Google Trend data is scraped on daily basis, it is needed to transform this data into 5-minute frequency data. Linear interpolation is the transformation method that is applied for this data, as it was argued by Moritz, Sardá, Bartz-Beielstein, Zaefferer, and Stork (2015) that linear interpolation is both computationally cheap and can reasonably well impute univariate time series data. This linear interpolation is performed by using the Pandas Package 'Interpolate' (Pandas PyData, 2021b).

Additionally, all the Twitter sentiment indicators are scraped on minute basis. As already explained in the last subsection Data, in Figure B.2 it is shown that those Twitter indicators vary much over time and do not seem to show any trend at all (like the Bitcoin price itself does show in Figure 3.1. For that reason, it is chosen to construct multiple Random Forest models with all the models containing different transformations of those social indicators to overcome this 'randomness' issue of the Twitter indicators. Then the model with the best performance in terms of profitability is chosen as the final model with social indicators. The transformations that are applied to these social indicators are Exponential Weighted Moving Average (EWMA) and including lagged social indicators. Holt (2004) has shown that EWMA can smoothen the randomness in time series data by putting declining weight on older data. In this research, the EWMA of all the social indicators (including the Google Trend indicator) are computed with the Pandas Package 'EWM' (Pandas PyData, 2021a). Next to this, the inclusion of lagged social indicators could also improve the profitability of a trading strategy (Garcia & Schweitzer, 2015). For that reason, it is also tested whether including the social features with a lag 1 (meaning a five minute lag) will improve the profitability of the trading strategy.

As the last step, all the financial features as well as all the social indicators are standardized before used as features in the Random Forest:

$$x_{standardized} = \frac{x - \hat{\mu}_i}{\hat{\sigma}_i},$$

with $\hat{\mu}_i$ denoting the mean of feature i and $\hat{\sigma}_i$ the standard deviation of that same feature.

Lastly, the performance of this model is evaluated by running a live algorithmic trading simulation. This simulation is a rolling, out-of-sample one like [Vo and Yost-Bremm \(2020\)](#) have used in their evaluations. This means that the simulation buys and sells Bitcoins based on the decision from the best performing machine learning model. If the price is expected to increase in the next period, Bitcoins will be bought or the position in Bitcoins will be kept. However, if the price is expected to decrease, the Bitcoin will be sold or the algorithm will decide to stay out of the market. The starting value that is used to buy Bitcoins in this simulation is \$1000. An extra factor that will be taken into account in this model is the trading cost on a cryptocurrency exchange. This means that only if the expected price increase is higher than the cost of trading on Bitcoins, the model will decide to buy or sell the Bitcoins. Note that these trading costs for trading on Binance are 0.1%. In this simulation, it is expected that the buy and sell model that is based on the model with parameters estimated from one week before that certain day will give the most profitable results ([Vo & Yost-Bremm, 2020](#)). Note that this means that the model will be updated every trading day. Finally, hypotheses [3.1](#) and [3.3](#) are then tested on the outcomes of this algorithmic trading simulation.

The performance of these Random Forest models is also compared by both the cumulative and the annualized strategy return. The cumulative strategy return is defined as follows ([J. Chen, 2021b](#)):

$$R_c = \prod_{t=1}^n (1 + r_t) - 1,$$

where r_t represents the daily return of day t and n the number of days in the trading simulation. Note that in this research, $n = 119 - 7 = 112$ days as the trading cannot be simulated before the end of the first week. Subsequently, the annualized strategy return is then defined as ([J. Chen, 2021a](#)):

$$R_a = (1 + R_c)^{\frac{365}{n}} - 1,$$

where R_c represents the cumulative strategy return and the number of days in the trading simulation is $n = 112$ days again.

Results

In this chapter, the results of the HFT algorithms for trading Bitcoins will be explained in detail. This means that the outcomes of these algorithms, their performance, and their statistical test results will be demonstrated.

The first HFT strategy that was constructed in this research is the Random Forest model with only the financial indicators. This strategy was then compared to the second HFT strategy, the Random Forest model with both the financial and the social indicators. As described in the last chapter [Methodology](#), to find the best performing model with both the financial and the social indicators, it was first needed to investigate the possible inclusion of the explained transformations of those social indicators in the Random Forest models. For that matter, five versions of the model with both financial and social indicators were created to test which model version would give the highest profitability in terms of both cumulative and annualized strategy return as well as in terms of annualized Sharpe ratio. The results are presented in Table [B.1](#) in Appendix [B](#). A description of the modeling process and the results of the five models is given as follows:

- A. **Version A** represents the most basic Random Forest model that contains the financial indicators as well as all the social indicators without any transformations. This was the first version of the trading strategy with social indicators. It was found that this strategy was not profitable and hence it was considered to include transformations of the social indicators in the model.
- B. **Version B** represents the Random Forest model that contains both financial indicators and the social indicators transformed by EWMA. It was found that this strategy made even more loss than version A and hence it was chosen to move on to models including lagged social indicators.
- C. **Version C** represents the Random Forest model including both financial indicators and lagged social indicators. This was the first one of these models that were computed that

rendered a profitable outcome. The feature importances of all the indicators used in this model are presented in Figure B.1. There it is shown that only the lagged 'Price' and lagged 'Google Trends' indicators were approximately as important as the financial indicators.

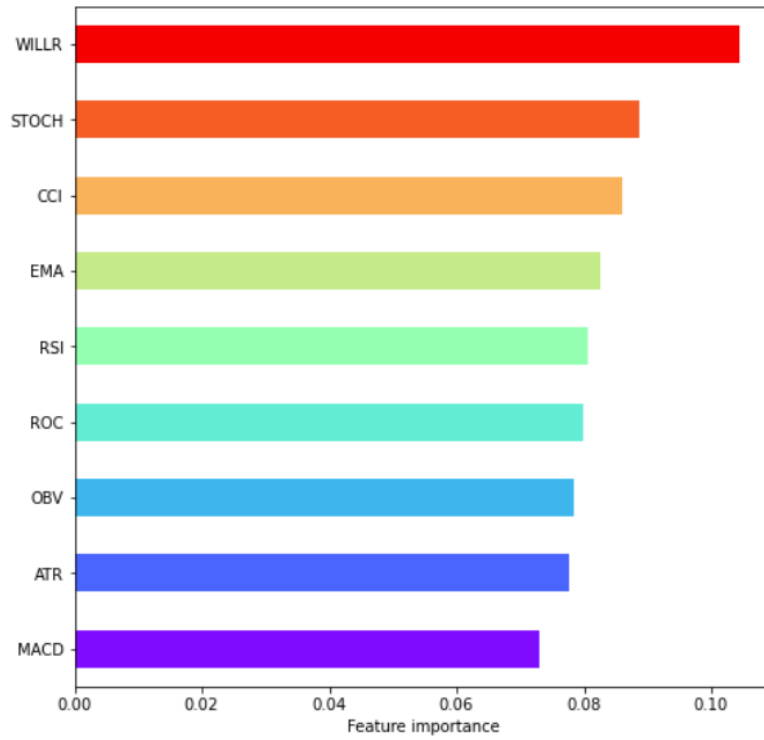
D. **Version D** represents the Random Forest model including the financial indicators, the social indicators, and the lagged social indicators. This model was tested to understand whether it was necessary to also include the non-lagged variables in the Random Forest. However, it was found that this model was less profitable than the model with only the lagged social indicators. Thus it was chosen to exclude the non-lagged social features from the final model.

E. The **Final Model** was found by combining the findings from Version C and Version D. This means that the final model represents the Random Forest model including the financial indicators and only the two lagged social indicators 'Price' and 'Google Trends', as these were the only important social indicators to take into account. It was found that this model was the most profitable among all the five versions in both cumulative and annualized strategy return as well as in annualized Sharpe ratio.

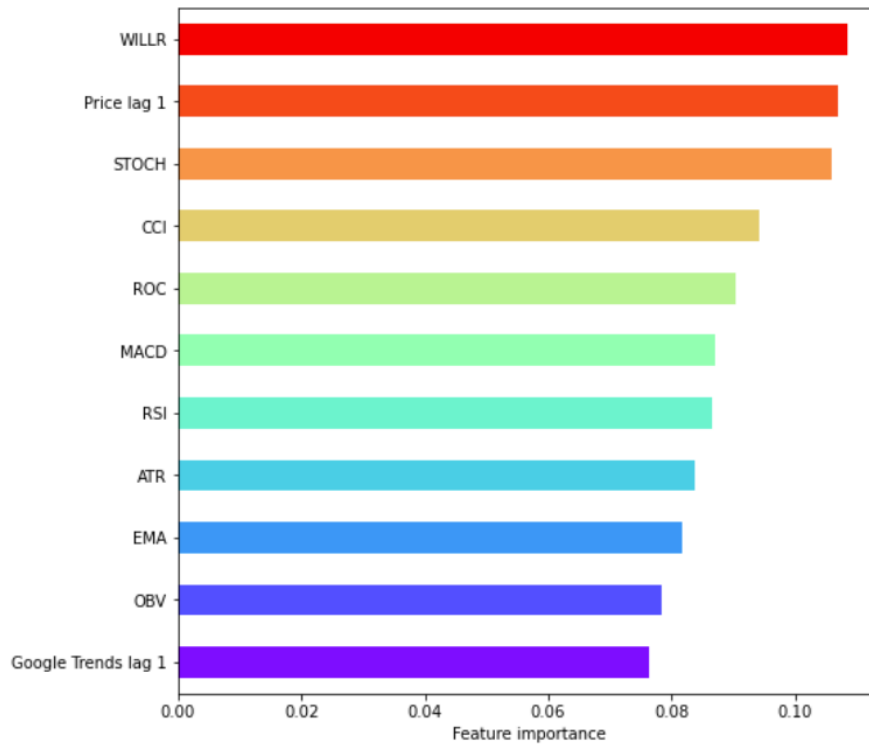
As an addition, the two lagged social indicators and their relation to the Bitcoin closing price are shown in Figures B.3 and B.4 in Appendix B. These two figures contain scatter plots of the two lagged indicators versus the closing prices with a linear regression line to better understand the relations between the variables on both axes. It is demonstrated that there is a slight negative between both the lagged social indicators and the Bitcoin closing prices.

The optimal parameter values for the Random Forest models were found using the SciKit Learn package 'GridSearchCV' (Scikit Learn, 2021). This package checks which parameter values from a grid of values give the most optimal results in a machine learning model using 10-fold cross-validation. Note that the optimal parameter values correspond with the model with the highest accuracy. In the end, it was found that a maximum number of 600 trees in the Random Forest was optimal in terms of model performance. The feature importances of both the final Random Forest models are reported in Figure 4.1. There it is demonstrated that for the model only including the financial indicators, all of those indicators apart from the *WILLR* are almost equally important in that Random Forest model, with the *WILLR* itself being the most important feature. However, for the model including the two lagged social indicators, it is shown that all indicators approximately have equal feature importance, although it should be noted that the lagged social indicator 'Price' has the second highest feature importance.

Both these final Random Forests models were then compared in a rolling, out-of-sample live algorithmic trading simulation as described in the last chapter Methodology. In this simulation, it was predicted whether the Bitcoin price would increase or decrease during 5-minute intervals on a certain day. These predictions were based on a Random Forest model that was



(a) Without the inclusion of social indicators.



(b) With the inclusion of the social indicators with highest feature importance.

Figure 4.1: The feature importances of the two Random Forest models with and without the social indicators.

	Model 1	Model 2	Model 3
	<i>Without socials</i>	<i>With socials</i>	<i>Buy&Hold</i>
<i>Average F1-score (%)</i>	51.6	57.6	-
<i>Average Accuracy (%)</i>	53.1	57.8	-
<i>Cumulative Strategy Return (%)</i>	2.55	4.37	82.8
<i>Annualized Strategy Return (%)</i>	5.10	14.9	288
<i>Annualized Sharpe Ratio</i>	1.92	2.02	1.04

Table 4.1: The results from the Random Forest trading simulations with and without social indicators.

trained with data from the week before that date. In this research, two separate trading simulations were constructed. The first simulation was based on Random Forests that were only trained on financial indicators, the second one on Random Forests that were trained on financial indicators and on the two lagged social indicators 'Price' and 'Google Trends' as explained before. Both models have taken the trading costs of 0.1% at Binance into account in the decision to buy/hold or to sell/leave. The outcomes of these two simulations can be found in Table 4.1, where these two models are described and compared to the Buy&Hold strategy. These results show that these outcomes are rather diverse: both the average accuracy and the average F1-score are higher in the model with the lagged social indicators than in the model without those. The same holds for both the cumulative and the annualized strategy return. The most outstanding result is the annualized strategy return of the model with the two lagged social indicators of 14.9%. However, the annualized Sharpe ratios of the two models are close to each other. This is caused by the finding that the standard deviation of the daily returns from the model with the two lagged social indicators is almost two times as high as the one from the model with only the financial indicators ($\sigma_1 = 0.000291$ versus $\sigma_2 = 0.000175$). From the table, it also becomes clear that the both Random Forest trading strategies are way less profitable in terms of both cumulative as well as annualized strategy return than the Buy&Hold strategy. However, because this Buy&Hold strategy also has a much higher standard deviation than the two algorithmic trading strategies ($\sigma_3 = 0.0472$), the annualized Sharpe ratio of this Buy&Hold strategy is the lowest one out of these three models.

To obtain a more in-depth understanding of the profitability of the two Random Forest trading strategies, Figure 4.2 is provided to observe the cumulative strategy return over the simulation period of January 1 to April 29. From this graph, it becomes clear that the simulation with the two lagged social indicators has higher returns during the whole simulation period than the simulation with only the financial indicators. The highest cumulative return of the strategy with the two lagged social indicators of almost 25% is found at March 26.

With the outcomes that are described in this subsection, the three hypotheses from the last subsection [Methodology](#) were tested. Firstly, hypothesis 3.1 was tested to understand whether the annualized Sharpe ratio of an HFT model would be significantly larger than the annualized Sharpe ratio of the Buy&Hold strategy. For this matter, it was chosen to test the

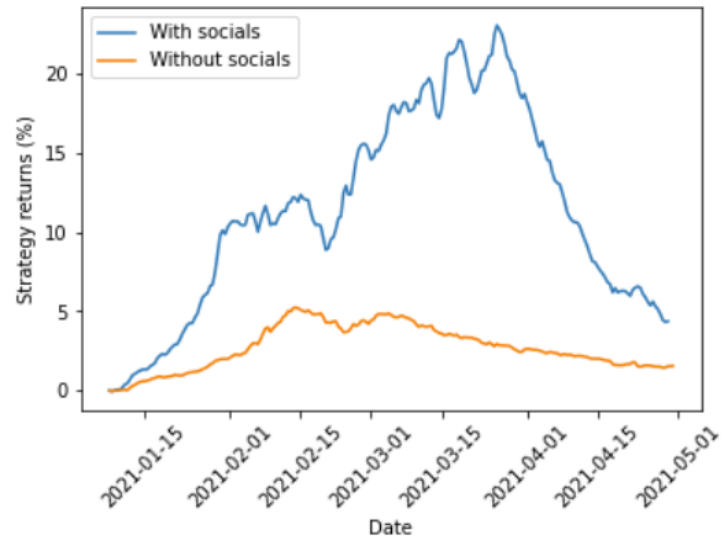


Figure 4.2: The strategy returns from simulation period for the models with and without the social features.

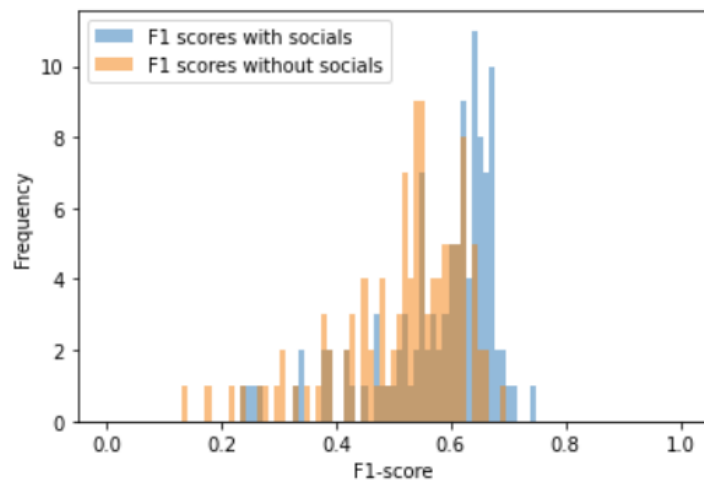


Figure 4.3: A histogram of the F1-scores of the Random Forest models with and without the social features added.

annualized Sharpe ratio of the Random Forest model with two lagged social indicators versus the annualized Sharpe ratio of the Buy&Hold strategy. The [Jobson and Korkie \(1981\)](#) test statistic $\hat{z}$ was therefore calculated to be equal to $\hat{z} = 0.616$. Given a significance level of 5%, the standard normal critical value for a one-sided test is given by 1.645. This means that H_0 of equal annualized Sharpe ratios is not rejected and thus that there is no significant result that the annualized Sharpe ratio of the HFT strategy is larger than the one of the B&H strategy.

Secondly, hypothesis [3.2](#) was tested to understand which of the two Random Forest models had a higher performance in terms of the F1-score. It was expected that the Random Forest model with the inclusion of the two lagged social features would have higher daily F1-scores. This was tested with the Wilcoxon signed ranks test and it was found that the p -value of this test is $p < 0.001$. This means that at a significance level of 5%, H_0 is rejected such that the daily F1-scores of the Random Forest model with the two lagged social indicators indeed are higher than the ones from the model without those indicators. This finding is also showed visually in [Figure 4.3](#), where the histograms of the daily F1-scores of both models are presented. These histograms show that the daily F1-scores are distributed similarly, but that the F1-scores of the model with social indicators are more centered around a higher mean than the ones of the model without those indicators.

Lastly, hypothesis [3.3](#) was tested to understand whether the annualized Sharpe ratio of the Random Forest model with the social indicators would be greater than the one of the models without those social indicators. This was again tested with the [Jobson and Korkie \(1981\)](#) test and the test statistic was found to be $\hat{z} = 0.16$. Comparing this again to the 5% critical one-sided test value 1.645 of the standard normal distribution, this means that the null hypothesis will not be rejected. Therefore, there is no evidence that the annualized Sharpe ratio of the Random Forest algorithm with the two lagged social features is bigger than the one without those social features.

Discussion and Conclusion

This research was focused on creating a profitable HFT algorithm that incorporated both financial and social features to train the strategy. In short, in the end two Random Forest models were trained on Bitcoin data from the exchange Binance. One of the models only considered financial indicators, the other model considered both financial and lagged social indicators in the training dataset. Firstly, it was found that the model with both the financial indicators and two lagged social indicators, namely 'Price' and 'Google Trends', had the highest performance in terms of profitability of all the Random Forest models including the social indicators that were considered. Then, it was concluded that the financial indicators were considered to be approximately equally as important as the two lagged social indicators, as all of the financial indicators had approximately equal feature importance as those two lagged social indicators. From statistical tests, it became clear that the model with the social indicators had a significantly higher F1-score than the model without those indicators. As was expected in the [Methodology](#), the accuracies and F1-scores of both the machine learning models were lower than the outcomes of papers that considered less recent data ([Borges & Neves, 2020](#); [Vo & Yost-Bremm, 2020](#)), as the Bitcoin market has moved towards efficiency ([Mnif et al., 2020](#)).

Subsequently, both the cumulative and the annualized strategy return of the model with the lagged social indicators over time were generally higher than the model without those indicators. After that, it was found that the model with the two lagged social features had a higher annualized Sharpe ratio, but that there was no significant result that the Sharpe ratio of that model was higher than the model without those social features. It has to be noted that this result is not in line with the results from [Garcia and Schweitzer \(2015\)](#) and [Z. Chen et al. \(2020\)](#) as covered in the [Theoretical Framework](#), as they showed that machine learning models that contained social indicators in the training dataset reached high profitability compared to other researches from the same trading period ([Borges & Neves, 2020](#); [Sebastião & Godinho, 2021](#)). Possibly, this could again have been caused by the finding that the market efficiency of

algorithmic Bitcoin trading has been surging these past few years ([Sensoy, 2019](#)). Therefore, the profits of algorithmic trading could have been stabilized by the increased market efficiency such that the differences between trading models with or without extra social features could not be able to generate higher profitability of the said strategy.

Lastly, while both the cumulative as well as the annualized strategy return of the Buy&Hold strategy were much higher than the ones of the two Random Forest models, the annualized Sharpe ratio of the Buy&Hold strategy was lower than the annualized Sharpe ratios of the two HFT algorithms. However, it has been shown that this annualized Sharpe ratio was not significantly lower than the ones from the two Random Forest models. Unlike past research ([Sebastião & Godinho, 2021](#); [Vo & Yost-Bremm, 2020](#)), this research has thus shown that algorithmic HFT-trading on the Bitcoin market does not generate a significantly higher Sharpe ratio, and thus not a higher risk-adjusted return, than a simple Buy&Hold strategy. This could again possibly be explained by the more efficient Bitcoin market that was considered in this research compared to the market from a few years ago.

As the focus of this research was on the profitability of HFT-strategies on trading Bitcoins while including social features, the only machine learning method that has been used is Random Forest. By the research described in the [Theoretical Framework](#), it was expected that this method would generate the most profitable strategies compared to the other methods ([Vo & Yost-Bremm, 2020](#)). However, it is not necessarily the case that Random Forest models can predict Bitcoin prices the most accurately in today's Bitcoin market as its market efficiency has grown over the last few years. Therefore, future research could focus on comparing multiple machine learning and possibly deep learning techniques with the outcomes of these Random Forest models to study which method would construct the most profitable HFT-strategy for trading Bitcoins. Next to this, in this research, a training period length in the trading simulation of one week was assumed to give the highest profitability ([Vo & Yost-Bremm, 2020](#)). However, in future research, it could be interesting to study whether this one week training period indeed gives the highest profitability or that a longer or shorter period would give better results.

Future research could also put even more focus on the method of including the social indicators in the trading strategy. For example, in this research, it was chosen to use linear interpolation to interpolate the Google Trend data on a 5-minute interval. It was outside the scope of this research to use a different method to estimate this data, but it could be interesting to see if the model would have a better performance if Midas Regression was used for instance ([Ghysels, Sinko, & Valkanov, 2007](#)). Furthermore, it also could be interesting to look at the inclusion of social indicators in algorithmic HFT-strategies for a longer period in time, for example for a few consecutive years in a row. Then it could be studied whether the difference in profitability between strategies with and without social indicators has indeed declined over time which has been argued before.

In sum, this research has focused on creating HFT-strategies for trading Bitcoins on cryptocurrency exchanges. While including lagged social features in that HFT-strategy causes a significantly higher F1-score, it does not invoke a significantly higher risk-adjusted return than an HFT-strategy without including those social features. Unlike past research, these two HFT-strategies do not seem to significantly outperform a simple Buy&Hold strategy, which creates an interesting topic for future research.

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Appendices

Indicators

A.1 Financial indicators

The nine financial indicators used in this paper are defined as follows ([Borges & Neves, 2020](#)).

1. **Relative Strength Index (*RSI*)**: *RSI* measures the movement of Bitcoin prices in terms of speed:

$$RSI = 100 - \frac{100}{1 + RS(p)}, \quad \text{where} \quad RS(p) = \frac{Ave(Gains)}{Ave(Losses)}.$$

This indicator is used to check if the Bitcoin is overbought ($RSI > 70$) or oversold ($RSI < 30$).

2. **Stochastic Oscillator (*STOCH*)**: *STOCH* is also a measurement of the price movement, now in relation to the current closing price C_k , the highest price H_p in a certain period p and the lowest price L_p :

$$STOCH = 100 \cdot \frac{C_k - L_p}{H_p - L_p}.$$

This indicator is used to check if the Bitcoin is overbought ($STOCH > 80$) or oversold ($STOCH < 20$).

3. **Rate of Change (*ROC*)**: *ROC* is another momentum measurement that calculates the percentual change of the current Bitcoin closing price C_k in a certain period p :

$$ROC(p) = 100 \cdot \frac{C_k - C_{k-p}}{C_{k-p}}.$$

4. **Exponential Moving Average (*EMA*)**: *EMA* refers to the method of smoothing short term oscillations by averaging new data:

$$EMA_k(p) = EMA_{k-1}(p) + \frac{2}{p+1} \left(C_k - EMA_{k-1}(p) \right).$$

Note that k refers to the current period and p to the number of periods to be averaged over.

5. **Moving Average Convergence-Divergence (MACD)**: *MACD* calculates a difference between a long-period *EMA* (usually period length of 26) and a short-period *EMA* (usually period length of 12):

$$MACD = EMA_{12} - EMA_{26}.$$

Then the signal line is calculated as follows:

$$Signal\ line = EMA_9(MACD).$$

If the *MACD* is above the signal line, it is an indication to buy Bitcoins and vice versa.

6. **Commodity Channel Index (CCI)**: *CCI* measures variation of the Bitcoin price compared to its statistical average:

$$CCI(n) = \frac{1}{0.015} \cdot \frac{TP_k - SMA_n(TP_k)}{\sigma_n(TP_k)}, \text{ where } TP_k = \frac{H_k + L_k + C_k}{3}.$$

Here *SMA* corresponds to simple moving averaging. A high *CCI* value means that prices are high compared to the average prices given previous time period n , so that it is thought to be oversold.

7. **On Balance Volume (OBV)**: *OBV* is an indicator that focuses on both volume V_k at current time k and current closing price C_k :

$$OBV_k = \begin{cases} OBV_{k-1} + V_k, & \text{if } C_k > C_{k-1} \\ OBV_{k-1} - V_k, & \text{if } C_k < C_{k-1} \\ OBV_{k-1}, & \text{if } C_k = C_{k-1}. \end{cases}$$

Note that the trend of the *OBV* line is important. If the *OBV* moves differently than the prices this shows a possible trend reversal coming.

8. **Average True Range (ATR)**: *ATR* is a measurement of the volatility of Bitcoin prices. The first step of calculation is to calculate the true range itself:

$$ATR_k = \max \left(H_k - L_k ; |H_k - C_{k-1}| ; |L_k - C_{k-1}| \right).$$

Then the *ATR* is the average of the true range:

$$ATR_k(n) = \frac{(k-1)ATR_{k-1} + ATR_k}{n}.$$

9. **Williams %R (*WILLR*)**: *WILLR* measures the current closing price relative to the high and low prices in a certain period p :

$$WILLR = -100 \cdot \frac{H_p - C_k}{H_p - L_p}.$$

When $WILLR < -80$, it is said to be oversold and when $WILLR > -20$, it is said to be overbought.

A.2 Social indicators

The social indicators from the Augmento API ([Augmento, 2021](#)) are given as follows.

1. **Bullish**: a Bullish sentiment corresponds to the believes of individuals that the Bitcoin prices are expected to rise in the coming period.
2. **Bearish**: a Bearish sentiment corresponds to the believes of individuals that the Bitcoin prices are expected to drop in the coming period.
3. **Optimistic**: an Optimistic sentiment corresponds to the believes of individuals that the Bitcoin market will grow in the future
4. **Pessimistic**: an Optimistic sentiment corresponds to the believes of individuals that the Bitcoin market will diminish in the future
5. **Cheap**: a Cheap sentiment corresponds to the believes of individuals that the Bitcoin prices are thought of to be cheap
6. **Expensive**: an Expensive sentiment corresponds to the believes of individuals that the Bitcoin prices are thought of to be expensive
7. **Buying**: a Buying sentiment corresponds to the believes of individuals that they want/are going to buy their Bitcoins
8. **Selling**: a Selling sentiment corresponds to the believes of individuals that they want/are going to sell their Bitcoins

This Twitter sentiment dataset contains minute-based data about the social indicators that are used in this research. The values in this dataset correspond to the number of tweets in a minute that belong to the sentiment of the social indicator. For example, a value for the social indicator 'Expensive' of 5 means that there were five tweets during one minute that fell into the Expensive sentiment.

Extra Graphs and Tables

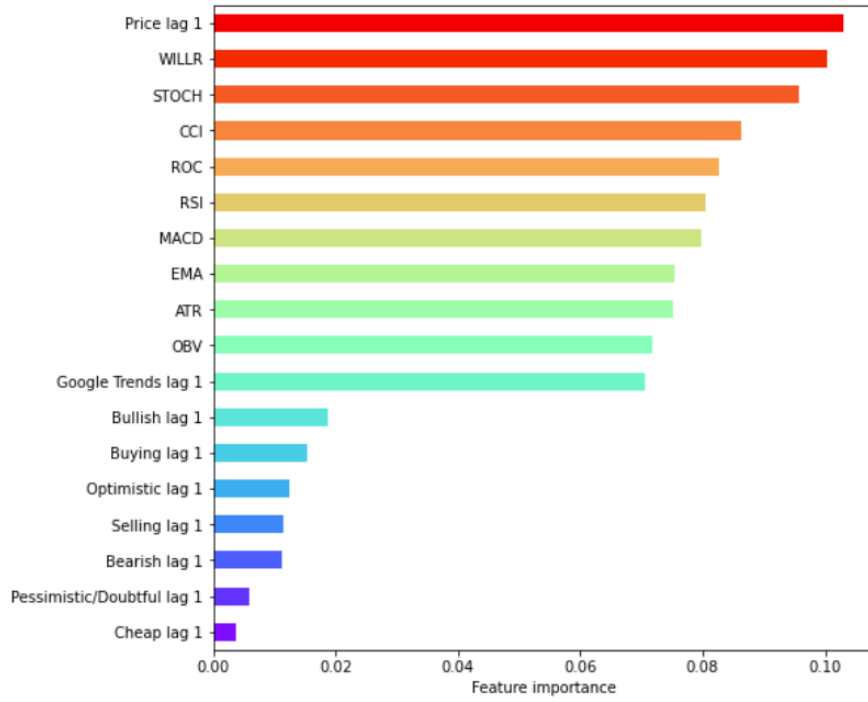


Figure B.1: The feature importance of the Random Forest model with the inclusion of all the lagged social indicators.

	Final Model	Version A	Version B	Version C	Version D
<i>Mean F1-score (%)</i>	57.6	52.2	50.4	57.3	56.8
<i>Mean Accuracy (%)</i>	57.8	53.0	52.5	57.8	57.4
<i>Cumulative Strategy Return (%)</i>	4.37	-0.99	-0.98	4.18	1.93
<i>Annualized Strategy Return (%)</i>	14.9	-2.81	-3.22	14.3	6.44
<i>Annualized Sharpe Ratio</i>	2.02	-2.11	-2.12	1.99	1.35

Table B.1: The results from the Random Forest models with the inclusion of different transformations on the social indicators.

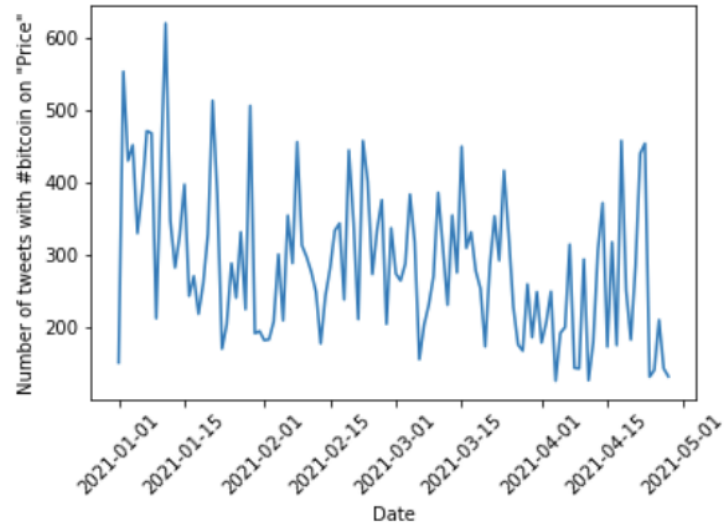


Figure B.2: The number of tweets per day about Bitcoins containing the word 'Price' in the period from January 1, 2021 to April 29, 2021.

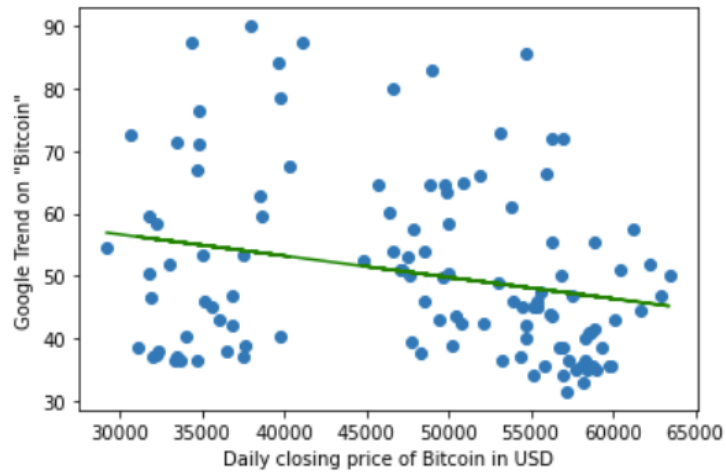


Figure B.3: A scatter plot with linear regression line of the lagged Google trend data on the word 'Bitcoin' and the daily Bitcoin closing prices on Binance for the period from January 1, 2021 to April 29, 2021.

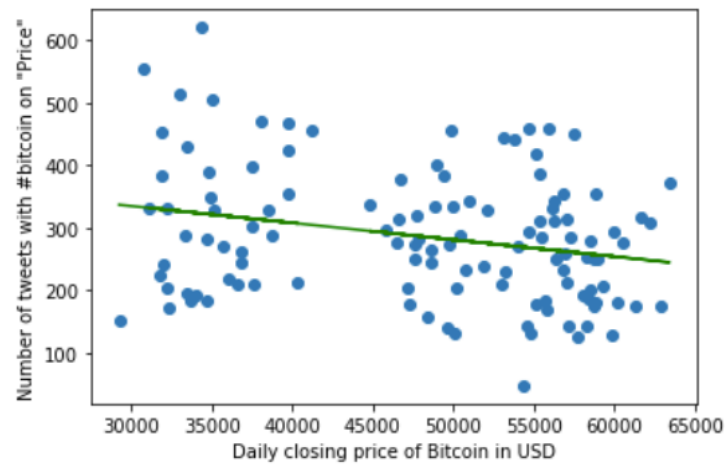


Figure B.4: A scatter plot with linear regression line of the lagged Twitter sentiment data on the hashtag '#bitcoin' about the indicator 'Price' and the daily Bitcoin closing prices on Binance for the period from January 1, 2021 to April 29, 2021.